

Parv Kapoor

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Education

Carnegie Mellon University

Ph.D. in Software Engineering | Advisors: Dr. Eunsuk Kang and Dr. Sebastian Scherer

Pittsburgh, PA

August 2026 (expected)

Carnegie Mellon University

Masters in Software Engineering | GPA: 4.03/4.00

Pittsburgh, PA

May 2025 (expected)

Manipal Institute of Technology

B.Tech. in Computer Science Engineering | GPA: 8.59/10

Manipal, India

August 2020

Skills

Programming Python, C, C++, CUDA programming, JAVA, MATLAB, Alloy, TLA+, MySQL, OpenCL

Tools and Libraries PyTorch, TensorFlow, ROS, Docker, Habitat, Isaac Gym, FSDP

Ph.D. Research

AirLab, Carnegie Mellon University

PI: Sebastian Scherer

Pittsburgh, PA

August 2021 - Ongoing

- Pretrained Learning from Demonstration (LfD) policies on real-world aerial trajectory data, and enhanced constraint satisfaction through on-line Monte Carlo Tree Search refinement achieving a 60 % improvement over traditional LfD techniques
- Developed a vision-based collision avoidance system for autonomous aircraft using control barrier functions, achieving 71 % improvement over baselines. Validated through digital twin simulations in Nvidia Isaac Sim and 70+ hours of in-field testing on Aurelia X6 drones
- Built a full stack deep learning library from scratch, with both CPU (C++) and GPU (CUDA) backends, enabling efficient automatic differentiation and custom RNNs and LLMs. Engineered implicit layers for end-to-end non-linear optimization

Software Design and Analysis Lab, Carnegie Mellon University

PI: Eunsuk Kang

Pittsburgh, PA

August 2021 - Ongoing

- Developing tools for Large Language Model alignment via automata constrained decoding for structured output generation
- Developed STLCG++, an attention masking based approach leveraging PyTorch and JAX for efficient sequence analysis, achieving an order of magnitude improvement over RNN-based methods
- Formulated a comprehensive theory and toolset for robustness analysis of safe reinforcement learning policies using stochastic optimisation
- Pioneered an optimization-driven trajectory synthesis framework using requirement decomposition for long horizon Task and Motion Planning. Achieved a 65 % reduction in solving time and a 51 % performance improvement over state-of-the-art planners

Experience

Scaled Foundations

Student Researcher (Part time) | PIs: Dr. Jonathan Huang and Dr. Ashish Kapoor

Seattle, WA

September 2024 - Ongoing

- Developed pipelines for distributed synthetic data generation, pretraining and fine tuning of robot foundation models
- Developed aligned robotics foundation models using novel fine tuning paradigms for navigation and instruction following
- Contributing to the development of the AirGen simulation platform for aerial, ground, and manipulator robots

Scaled Foundations

Research Intern | PIs: Dr. Sai Vemprala and Dr. Ashish Kapoor

Seattle, WA

May 2024 - August 24

- Developed a novel technique for safe trajectory generation extending state-of-the-art autoregressive robotics pretraining architecture, outperforming baselines by 74.3%.
- Built a foundation model field deployment pipeline for diverse robots (Unitree Go2, DJI Robomaster) using domain adaptation.
- Integrated multimodal foundation models into the GRID platform, developed and deployed skills chaining the models.

Verimag, Université Grenoble Alpes

Research Engineer | PI: Dr. Thao Dang

Grenoble, France (Remote)

January 2021 - August 2021

- Developed input stimulus generation theory using timed automata for autonomous system validation
- Evaluated these techniques within the SUMO simulation environment for applications in autonomous vehicles

Cyber Physical Systems Lab, University of Southern California

Research Intern | PI: Dr. Jyotirmoy Vinay Deshmukh

Los Angeles, U.S.A.

January 2020 - January 2021

- Developed novel model-based reinforcement learning algorithms for safe policy training from signal temporal logic specifications
- Implemented efficient model-free algorithms (TRPO, A3C, PPO) in PyTorch with unique STL-based reward design
- Achieved 82% higher specification satisfaction compared to baseline RL policies
- Engineered in-house simulation environments for algorithm benchmarking employing CARLA, AirSim, and Gazebo

Visual Computing Group, Cardiff University

Research Intern | PI: Dr. David Marshall

Cardiff, U.K.

May 2019 - July 2019

- Constructed a safe trajectory prediction system for visually impaired individuals using ZED stereo camera
- Implemented and trained a 2-stream CNN in TensorFlow on human walking data for forecasting ego agent camera movement
- Improved CNN accuracy in low-data regimes through neuro-inspired data augmentation

RapidQube Digital solutions Pvt. Ltd.

Research Intern

Mumbai, India

May 2018 - July 2018

- Created an accident prediction system leveraging convolutional neural networks and object tracking algorithms (YOLOv3)
- Implemented depth prediction Residual CNNs alongside YOLO v3 in Tensorflow to classify nearby drivers' speed profiles with 300 ms latency

Preprints and Publications

Pretrained Embeddings as a Behavior Specification Mechanism

P. Kapoor, Andy Hammer, Ashish Kapoor, Karen Leung, Eunsuk Kang

- Under submission [\[arxiv\]](#)

STLCG++: A Masking Approach for Differentiable Signal Temporal Logic Specification

P. Kapoor, Kazuki Mizuta, Eunsuk Kang, Karen Leung

- Under submission [\[arxiv\]](#)

Logically Constrained Robotics Transformers for Enhanced Perception Action Planning

P. Kapoor, Sai Vemprala, Ashish Kapoor

- Robotics Science and Systems (RSS): Towards Safe Autonomy 2024 [\[arxiv\]](#)

Constrained LTL Specification Learning from Examples

C. Zhang, P. Kapoor, I. Dardik, A. Cui, R.M. Goes, D. Garlan, E. Kang

- IEEE International Conference on Software Engineering (ICSE) 2025 [\[arxiv\]](#)

Analyzing Tolerance of Reinforcement Learning Controllers against Deviations in Cyber Physical Systems

C. Zhang*, P. Kapoor*, R.M. Goes, D. Garlan, E. Kang, A. Ganlath, S. Mishra, N. Ammar

- Formal Methods (FM) 2024 [\[arxiv\]](#)

Safe Planning through Incremental Decomposition of Signal Temporal Logic

P. Kapoor, R.M. Goes and E. Kang

- Nasa Formal Methods (NFM) 2024 [\[arxiv\]](#)

ViSafe: Vision-enabled Safety for High-speed Detection and Avoidance

P. Kapoor, I. Higgins, N. V. Keetha, J. Patrikar I. Cisneros, Z. Ye, Y. He, Y. Hu, S. Scherer

- Under Review

FoundLoc: Vision-based Onboard Aerial Localization in the Wild

Y. He, I. Cisneros, N. V. Keetha, J. Patrikar, Z. Ye, I. Higgins, Y. Hu, P. Kapoor, S. Scherer

- Under Review [\[arxiv\]](#)

Follow The Rules: Online Signal Temporal Logic Tree Search for Guided Imitation Learning in Stochastic Domains

J. Patrikar, J. Aloor, P. Kapoor, S. Scherer and J. Oh

- IEEE International Conference on Robotics and Automation (ICRA) 2023 [\[arxiv\]](#)

Challenges in Close-Proximity Safe and Seamless Operation of Manned and Unmanned Aircraft in Shared Airspace

J. Patrikar, J. Dantas, S. Ghosh, P. Kapoor et al

- IEEE International Conference on Robotics and Automation (ICRA) 2022 [\[arxiv\]](#)

Model-based Reinforcement Learning from Signal Temporal Logic Specifications

P. Kapoor, A. Balakrishnan, J. V. Deshmukh

- 2020 [\[arxiv\]](#)